

IGBOYA APPLICATION SOFTWARE (IAS)

Mathematics Performance Test (MPT)

Name of School:

Present Class:

Gender:

School Ownership: Public () Private ()

Instruction: Answer All Questions

1. Simplify: $(3x^2y) \times (2xy^3)$
 - A. $5x^3y^4$
 - B. $6x^3y^4$
 - C. $6x^2y^3$
 - D. $5x^2y^3$
2. If $2x - 3 = 7$, find the value of x.
 - A. 2
 - B. 4
 - C. 5
 - D. 6
3. Factorize completely: $x^2 - 9$.
 - A. $(x - 9)(x + 1)$
 - B. $(x - 3)(x + 3)$
 - C. $(x - 1)(x + 9)$
 - D. $(x + 9)(x + 3)$
4. Simplify: $(4x^2y) / (2xy^2)$
 - A. $2x/y$
 - B. $2y/x$
 - C. x^2/y^2
 - D. x/y
5. Solve for x: $5x + 7 = 2x + 13$
 - A. $x = 1$
 - B. $x = 2$
 - C. $x = 3$
 - D. $x = 4$
6. Expand: $(x + 2)(x - 3)$
 - A. $x^2 - x - 6$
 - B. $x^2 + x - 6$
 - C. $x^2 + 5x - 6$
 - D. $x^2 - 5x + 6$
7. If $x = 2$ and $y = -3$, find the value of $x^2 + 2xy + y^2$.
 - A. 1
 - B. 0
 - C. 2
 - D. -1
8. Make x the subject of the formula: $y = 3x + 5$
 - A. $x = (y - 5) / 3$
 - B. $x = (y + 5) / 3$
 - C. $x = 3y - 5$

- D. $x = 3y + 5$
9. The roots of the equation $x^2 - 5x + 6 = 0$ are:
- A. 2 and 4
 - B. 2 and 3
 - C. 1 and 6
 - D. 3 and 5
10. Simplify: $(x + y)^2 - (x - y)^2$.
- A. $2xy$
 - B. $4xy$
 - C. $x^2 + y^2$
 - D. $2x^2y^2$
11. The sum of the interior angles of a triangle is
- A. 90°
 - B. 120°
 - C. 180°
 - D. 360°
12. The exterior angle of a regular hexagon is
- A. 45°
 - B. 60°
 - C. 90°
 - D. 120°
13. In a right-angled triangle, if one of the acute angles is 30° , what is the other acute angle?
- A. 30°
 - B. 45°
 - C. 60°
 - D. 90°
14. A triangle with sides 3 cm, 4 cm, and 5 cm is
- A. Isosceles
 - B. Equilateral
 - C. Right-angled
 - D. Scalene
15. The sum of the angles of a quadrilateral is
- A. 180°
 - B. 270°
 - C. 360°
 - D. 540°
16. The diagonals of a rhombus
- A. Are equal
 - B. Bisect each other at right angles
 - C. Are parallel
 - D. Are unequal and do not bisect
17. The area of a triangle is given by
- A. $\frac{1}{2} \times \text{base} \times \text{height}$
 - B. $\text{base} \times \text{height}$
 - C. $2 \times \text{base} \times \text{height}$
 - D. $\text{base} + \text{height}$
18. The radius of a circle is 7 cm. Find its circumference. (Take $\pi = 22/7$)
- A. 14 cm
 - B. 21 cm
 - C. 44 cm
 - D. 49 cm

19. In a circle, an angle subtended by a diameter on the circumference is
- A. 30°
 - B. 45°
 - C. 60°
 - D. 90°
20. The length, breadth, and height of a rectangular box are 8 cm, 5 cm, and 3 cm respectively. Find its total surface area.
- A. 94 cm^2
 - B. 118 cm^2
 - C. 158 cm^2
 - D. 180 cm^2
21. If $y = x^2$, find dy/dx .
- A. x
 - B. $2x$
 - C. x^2
 - D. $3x$
22. If $y = 3x^3$, then $dy/dx =$
- A. $6x$
 - B. $9x^2$
 - C. $3x^2$
 - D. $12x$
23. If $y = 5x^4 + 2x$, find dy/dx .
- A. $20x^3 + 2$
 - B. $10x^2 + 2x$
 - C. $5x^3 + 2x$
 - D. $20x + 2$
24. If $y = 7x^2 - 4x + 3$, then $dy/dx =$
- A. $14x - 4$
 - B. $14x + 4$
 - C. $7x - 4$
 - D. $14x + 3$
25. If $y = 1/x$, find dy/dx .
- A. x
 - B. $-x^{-2}$
 - C. x^{-2}
 - D. $-x^2$
26. Differentiate $y = 4x^3 - 2x^2 + 6x - 5$.
- A. $12x^2 - 4x + 6$
 - B. $8x^2 - 4x + 5$
 - C. $12x^3 - 4x^2 + 6x$
 - D. $6x^2 + 4x + 6$
27. If $y = \sin x$, then $dy/dx =$
- A. $-\sin x$
 - B. $\cos x$
 - C. $-\cos x$
 - D. $\tan x$
28. The integral of x^2 with respect to x is
- A. $(x^3 / 3) + C$
 - B. $3x^2 + C$
 - C. $2x + C$
 - D. $x^3 + C$

29. If $dy/dx = 6x$, find y .
- A. $3x^2 + C$
 - B. $6x^2 + C$
 - C. $x^2 + C$
 - D. $12x + C$
30. If $y = e^x$, find dy/dx .
- A. 1
 - B. $x e^{(x-1)}$
 - C. e^x
 - D. xe^x
31. The value of $\sin 30^\circ$ is
- A. $\sqrt{3}/2$
 - B. $1/2$
 - C. $1/\sqrt{2}$
 - D. 1
32. Evaluate $\tan 45^\circ$.
- A. 0
 - B. 1
 - C. $\sqrt{3}$
 - D. $1/\sqrt{3}$
33. If $\cos A = 0.6$, find $\sin A$.
- A. 0.8
 - B. 0.4
 - C. 0.5
 - D. 1.0
34. Simplify $\sin 60^\circ / \cos 30^\circ$.
- A. 1
 - B. $\sqrt{3}$
 - C. $1/\sqrt{3}$
 - D. 2
35. Find the value of $\tan \theta$ if $\sin \theta = 4/5$ and $\cos \theta = 3/5$.
- A. $3/4$
 - B. $4/3$
 - C. $5/4$
 - D. $4/5$
36. If $\tan A = 1/\sqrt{3}$, then $A =$
- A. 30°
 - B. 45°
 - C. 60°
 - D. 90°
37. If $\sin \theta = 0.8$, find θ (to the nearest degree).
- A. 30°
 - B. 37°
 - C. 45°
 - D. 53°
38. Simplify $1 / \sec \theta$.
- A. $\sin \theta$
 - B. $\tan \theta$
 - C. $\cos \theta$
 - D. $\cot \theta$
39. If $\tan \theta = 5/12$, find $\sin \theta$.

- A. $5/13$
 - B. $12/13$
 - C. $13/5$
 - D. $5/12$
40. If $\sin \theta = 1/2$, then $\theta =$
- A. 30°
 - B. 45°
 - C. 60°
 - D. 90°
41. Find the median of the following data: 2, 5, 3, 8, 7, 9, 4.
- A. 4
 - B. 5
 - C. 6
 - D. 7
42. The mode of the data 3, 4, 5, 4, 6, 7, 4, 8 is
- A. 4
 - B. 5
 - C. 6
 - D. 7
43. If the mean of five numbers is 12, find their total sum.
- A. 50
 - B. 55
 - C. 60
 - D. 65
44. A student scored 8, 10, 6, and 12 marks in four tests. What must he score in the fifth test to have a mean of 10?
- A. 10
 - B. 12
 - C. 14
 - D. 16
45. The range of the data 2, 4, 6, 10, 8 is
- A. 6
 - B. 7
 - C. 8
 - D. 10
46. For the data set 5, 7, 9, 11, 13, the variance is
- A. 4
 - B. 5
 - C. 6
 - D. 8
47. Which of the following is not a measure of central tendency?
- A. Mean
 - B. Median
 - C. Range
 - D. Mode
48. In a frequency distribution, the sum of frequencies is 50 and the sum of all observations is 250. Find the mean.
- A. 4
 - B. 5
 - C. 6
 - D. 7

49. The mode of a distribution is 16 and the mean is 14. If the data is moderately skewed, the median is approximately
- A. 15
 - B. 14
 - C. 16
 - D. 13
50. A group of students recorded the following ages (in years): 13, 14, 13, 15, 14, 14, 13, 15. What is the median age?
- A. 13.5
 - B. 14
 - C. 14.5
 - D. 15